

A HAMILTONIAN CUBIC GRAPH WITH NO CYCLE OF LENGTH $n - 1$

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ABSTRACT. Gordon Royle asked whether some non-bipartite Hamiltonian cubic graph on n vertices has no cycle of length $n - 1$, and reported that a computer found none on up to 24 vertices. Replacing each vertex of K_4 by a copy of $K_{2,3}$ gives one on 20 vertices. The proof is a two-line count of cycle edges in one block, and it applies to any Hamiltonian non-bipartite cubic graph in place of K_4 . For the 20-vertex graph every statement is formally verified in Lean.

1. THE QUESTION

A cycle of length n in a graph on n vertices is Hamiltonian. In a bipartite graph every cycle has even length, so a Hamiltonian bipartite graph on n vertices has no cycle of length $n - 1$; the cube is an example. In 2017 Royle asked on MathOverflow [1] whether a cubic graph can be Hamiltonian, not bipartite, and still have no cycle of length $n - 1$, and reported that his computer found none on up to 24 vertices. The question has had no answer there.

Theorem 1. *Let G be the graph obtained from K_4 by replacing each vertex by a copy of $K_{2,3}$, the three edges at that vertex being attached to the three vertices of the larger class. Then G is a cubic graph on 20 vertices that is Hamiltonian, is not bipartite, and has no cycle of length 19.*

2. THE CONSTRUCTION

Start from any cubic graph K . Replace each vertex v by a block X_v , a copy of $K_{2,3}$. Call its two vertices of degree three the *inner* vertices and its three vertices of degree two the *attachments*, and join the three attachments of X_v to the three edges of K at v , one each. Every vertex now has degree three, and the graph has $5|V(K)|$ vertices. With $K = K_4$ this is the graph of Theorem 1, drawn in Figure 1.

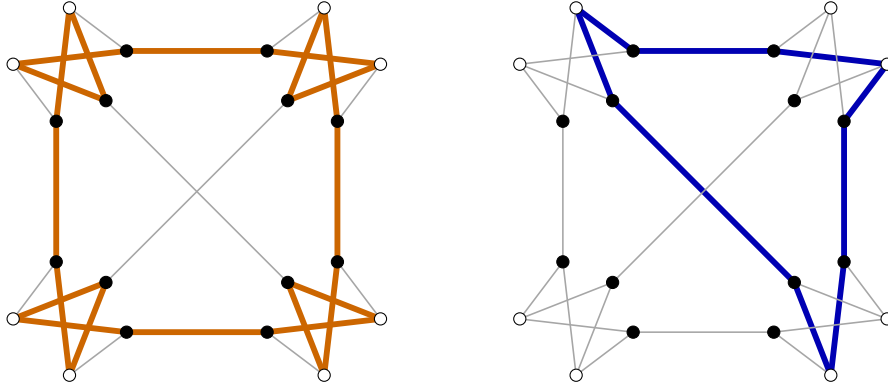


FIGURE 1. K_4 with each vertex replaced by $K_{2,3}$: inner vertices white, attachments black. Left, a Hamiltonian cycle. Right, a cycle of length nine.

Hamiltonian. Any two attachments p, p' of a block are the ends of a path through all five of its vertices: p , an inner vertex, the third attachment, the other inner vertex, p' . Following a Hamiltonian cycle of K and crossing each block by such a path gives a Hamiltonian cycle of the new graph (Figure 1, left).

Not bipartite. Two attachments of a block are also joined by a path of length two through an inner vertex. Following a triangle of K this way gives a cycle of length $3 \cdot 2 + 3 = 9$ (Figure 1, right).

3. NO CYCLE OF LENGTH $n - 1$

Lemma 2. *If K is any cubic graph, the graph built from it has no cycle of length $n - 1$.*

Proof. Suppose a cycle C passes through every vertex except one, say m , which lies in the block X . The other four vertices of X lie on C and each has two edges of C . Count these eight edge-ends. An edge of C with both ends in X is counted twice and an edge of C leaving X once, so if C has i edges inside X and e edges leaving it,

$$2i + e = 8.$$

Every edge inside X joins an inner vertex to an attachment, and an inner vertex has no neighbours outside X . So each inner vertex on C contributes both of its C -edges to i , and no edge inside X is counted from two inner vertices: i is twice the number of inner vertices on C .

If m is an inner vertex, one inner vertex is on C , so $i = 2$ and $e = 4$. But only three edges leave X , one at each attachment. If m is an attachment, both inner vertices are on C , so $i = 4$ and $e = 0$. Then C never leaves X and has at most five vertices, while $n - 1 \geq 19$. Either way we reach a contradiction. $\square$

Theorem 1 follows. The same argument applies to every bipartite block attached by three edges whose larger class carries the attachments; and replacing K_4 by any Hamiltonian cubic graph that is not bipartite, such as the triangular prism, gives further examples, on 30 vertices for the prism.

Remark. The theorem disagrees with the search reported in the question, and we do not know how that search was restricted. Our graph has girth four and three-edge cuts that separate a block from the rest, so it lies outside the classes of cubic graphs most often searched, such as those of girth five or those that are cyclically 4-edge-connected. Whether an example exists in those classes, and the least number of vertices of any example, we leave open.

VERIFICATION AND USE OF AI TOOLS

For the 20-vertex graph every statement of Theorem 1 is formally verified in Lean 4 with Mathlib (folder `lean`, standard axioms only). The theorem about cycles is proved for every cycle, in Mathlib's sense, by the count of Lemma 2; the finite facts about the fixed graph (its degrees, its neighbourhoods and the two cycles of Figure 1) are checked by evaluation. Six deliberately wrong variants of the formal statements fail to compile. Independently, the program `verify/check_cycles.py` builds the graph separately and confirms by exhaustive search that deleting any one vertex leaves a non-Hamiltonian graph, with the cube and the prism as controls; it finds the same for the 30-vertex example. The author used an AI tool (Claude, Anthropic) to find the example and write the formal proof, and is responsible for the content.

REFERENCES

- [1] G. Royle, Is there a non-bipartite hamiltonian cubic graph on n vertices with no $(n - 1)$ -cycle?, MathOverflow question 263706 (2017), <https://mathoverflow.net/q/263706>, accessed 9 October 2026.